Mariia Legenkaia

PhD in statistical physics and computational neuroscience

Contact

37 avenue Jean Jaurès
94110 Arcueil, France
+33 (777) 97 29 12
legenkai@bio.ens.psl.eu

Languages

Russian - native
English - fluent
French - fluent

Skills

Analytical methods

- High-dimensional statistics and inference
- Advanced Linear Algebra & Random matrix theory
- Replica method, saddle-point analysis
- High-dimensional covariance estimation

Machine learning & data analysis

- Dimensionality reduction (PCA, latent models)
- Uncertainty quantification
- Time series modeling with noise & heterogeneity

Programming

- Python (NumPy, SciPy, Pandas, JAX, PyTorch)
- Matlab
- Data visualization & simulation

Professional experience

2025 Research Scientist *École Normale Supérieure, Paris*
- Developed a high-performance library (JAX/NumPy) to quantify data requirements for reliable dimensionality reduction (PCA). Engineered benchmarking pipelines to detect signal-to-noise phase transitions, ensuring robust latent-structure recovery in high-dimensional datasets.
- Applied the framework to real neural recordings to estimate data requirements for reliable latent-dynamics recovery.

2021–2024 PhD in Statistical Physics & Computational Neuroscience*École Normale Supérieure, Paris*
- Developed a theoretical framework to predict PCA recovery accuracy in high-dimensional noisy data.
- Extended the spiked covariance model to heterogeneous noise and temporal correlations.
- Derived analytical error bounds using random matrix theory and replica methods.

Internships

2021 École Normale Supérieure, Paris *Paris, France*
M2 Intern
- Performed data-driven modelling of neural activity in sensory cortices using Restricted Boltzmann Machines

Education

2019–2021 Master 1 & 2 - theoretical physics *École Normale Supérieure, Paris*

2015–2019 Bachelor of Physics *Saint Petersburg National Research Academic University Of The Russian Academy Of Sciences*

Conferences and schools

2023 Summer school on Mathematical Methods in Computational Neuroscience *Fred Kavli Science Center, Norway*
- Attended as a student. Main topics: Information theory, liear and non-linear dimensionality reduction for neuroscience.

2022 FENS Forum *Paris, France*
- Participated in a poster session: Statistical limitations in the reconstruction of low-dimensional neural trajectories from high-dimensional activity recordings

2015-2018 Summer Camp of Saint Petersburg School for Young Mathematicians *Saint-Petersburg, Russia*
- Gave lectures in mathematics, conducted exercise sessions in mathematics and physics

Publications

2025 Uncertainties in signal recovery from heterogeneous and convoluted time series with principal component analysis
M. Legenkaia, L. Bourdieu, R. Monasson
https://doi.org/10.1103/PhysRevE.111.044314